



With better hardware, it is now possible to simulate smoke, dust storms and other particle effects

Moreover, weather changes such as fog, rain, dust-storms and day and night time cycles make the world of *Morrowind* much more believable and immersive.

Particle by particle

But *Morrowind* is just the herald of things to come. Particle effects are coming of age, adding realism to what was hitherto a very drab environment. Mark Rein, Vice-President of Epic Games Inc talks about the Unreal Engine: "A dynamic particle system is used to simulate dusty areas illuminated by light beaming through windows; and the dust moves around the air realistically as it gets disturbed by moving objects." Westka Interactive's *The Y Project* (a game combining action-adventure with role-playing elements, based in a science fiction/fantasy setting) uses this system to allow for impressive looking fires, sparks, smoke and foaming water. The particles are all tracked by the engine and since they react to objects passing through them, "when you fire a missile into a cloud of smoke, the smoke realistically dissipates", or if an invisible monster runs across a room filled with smoke, you will be able to see where it is by watching the smoke swirl as the creature moves through it.

Mark adds, "Another stunning effect is projective textures. They are used to create, among other things, soft-edge, ultra-detailed, real-time shadows, that cast on the world as well as the objects and characters within it. The intricate shadows not only render over the characters and the environments but also over the particles; as those particles are blown around by things flying through the air!" Immersive and interactive, isn't that what a game is all about?

In *The Y Project*, 30 per cent of the game occurs outdoors, where the particle system adds its magic touch. Clouds moving across the sky send real-time shadows creeping across the ground. The same system is used to create ripples and waves in water as you wade through it.



Regardless of the type of game, special effects are making a big difference in creating believable fantasies. A treat for F1 fans everywhere is Electronic Arts' *F1 2002*—realistic cars, accurately modelled real-world tracks, and anatomically correct models. These are just a few of the details that put you inside the world of F1 racing. The real treat though, is the simulation of weather, with rain battering down on your car, while raindrops settle on the windscreen and refract your view of the world, or the burned rubber that swirls from screeching tyres and dissipates languidly into the air.

But does creating a world in detail ensure immersion? Take a look around you and you will see the huge impact light has on our world. Even its absence adds a layer to realism. Just like the developers of the Unreal Engine, this truth hasn't escaped the guys behind *DOOM III*. For starters, all models will 'self-shadow', which is just programmer-speak for saying that parts of your body will cast shadows on other parts of your body. It doesn't sound like a lot, but it all adds up in

the end. Secondly, models will cast shadows on other objects in their surrounding, so you might see a large shadow envelope everything in your sight before seeing the monster that is attached to it.

It's in the details

You must have experienced moments in a game when you knew something didn't feel right, but you could not be sure what. "Gee, isn't that guy I just shot supposed to fall down the stairs?" or maybe even "he sure looks funny with his pants on fire". You have been a victim of physicsitis. It is a disease that is transmitted to gamers when the game programmers have taken shortcuts to building their physics engine. Thanks largely to faster processors, game developers now have the power to go all the way with physics. People will fall down stairs realistically when shot and pants will look dandy when on fire!

Mark Rein, Epic Games, says, "Karma is a rigid-body physics engine that has been integrated into the Unreal Engine and [is] available to [appropriate] Unreal Engine licensees. It allows you to simulate solid objects such as crates, tires or bones, as well as different joints, motors or springs between objects. Using these simple building blocks, complicated systems can be easily built that will behave according to the laws of physics. This can be anything from the classic stack of crates, to swinging lights or doors that can be shot off their hinges, up to 18-wheeler trucks, or characters falling down stairs like a rag-doll when they die." So you can shoot at overhead lights, which will make them

War Games

Bohemia Interactive Studios' *Operation Flashpoint* seems to have attracted the attention of the US Army. The developers are building a 'Military Grade Program' based on *Operation Flashpoint* technology called VBS1 (Virtual Battlefield System 1). The system will initially be used to train US marines, and has attracted other military organisations across the world for the purpose of training their troops.

While the VBS1 system is top-secret stuff, another project in the works is public domain. *America's Army: Operations* and its companion game, *America's Army: Soldiers*, are both being developed by the US Army in the hope of persuading the youth to join the forces. Using the latest Unreal Engine, *AA: Operations* will be an online shooter



Putting the latest Unreal Engine through its paces is the US Army with its upcoming games *America's Army: Operations* and *America's Army: Soldiers*

along the lines of *Counter-Strike*, while, *AA: Soldiers* will be more of a shooter-cum-RPG experience, charting your progress through the army.